

SHILPA J

Research Scientist | Enzyme R&D | Bioassays | Process & Analytical



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Bangalore, India

PROFILE

Research Scientist with over two years of industry experience in enzyme R&D, bioassay development and quality aligned laboratory environments. Experienced in enzyme assay design and optimization, protein expression, downstream processing and lab to bench scale-up. Strong exposure to documentation, method consistency, data integrity and SOP driven workflows supporting R&D, bioassay validation and QC functions.

EXPERIENCE

Research Scientist

KCAT Enzymatic Pvt. Ltd., Bangalore

Feb 2023 – Jul 2025

Optimized enzyme activity, stability and yield through systematic assay design across varying pH, temperature and substrate conditions, supporting improved process performance. Led laboratory to bench scale-up studies (up to 10 - 50 g substrate scale) ensuring process stability, consistent product quality and efficient downstream processing. Performed TLC screening and applied flow biocatalysis techniques for reaction monitoring, accelerating experimental workflows. Executed downstream processing for product separation and purification with strict quality control and documentation adherence. Maintained comprehensive experimental documentation ensuring data integrity, traceability, and consistency. Supported quality aligned laboratory practices including SOP adherence, raw data verification and traceable documentation to ensure audit readiness and data integrity.

INTERNSHIPS

Research Intern

KCAT Enzymatic Pvt. Ltd., Bangalore

Oct 2022 – Feb 2023

Optimized protein expression processes to achieve high-yield enzyme production for industrial applications. Gained hands-on expertise in enzyme assay development and biochemical characterization of proteins.

Project Intern

Scientific Industrial and Research Centre Pvt. Ltd., Bangalore

Jul 2021 – Sep 2021

Conducted master's thesis research on microbial production of protease via submerged fermentation. Performed bacterial isolation, gram staining, biochemical characterization, culture optimization, enzyme concentration (dialysis), kinetics studies and application analysis.

EDUCATION

M.Sc. in Biochemistry (2019 -2021) - Oxford College of Science, Bangalore

B.Sc. in PCM (2016 - 2019) - APS College of Arts and Science, Bangalore

PUC - PCMB (2014 - 2016) - NSVK College of Science, Bangalore

SSLC (2014) - Cauvery Vaidyanathan English High School, Bangalore

SKILLS

Analytical and Quality Techniques:

TLC Screening, SDS PAGE, Agarose Gel Electrophoresis, Basic HPLC Operation, Quality Control, Method Documentation, Data Integrity.

Enzyme R&D and Bioassays:

Enzyme Assay Design, Optimization and Validation, Activity and Kinetics Studies.

Process Development and Scale Up:

Exposure to Upstream and Downstream Processing, Lab to Bench Scale Up, Protein Expression and Purification.

Data and Collaboration:

Data Analysis, Technical Reporting, Cross Functional Collaboration.

Compliance Exposure:

Familiar with SOP driven workflows, controlled documentation practices, raw data traceability, and quality aligned laboratory operations.

CO-CURRICULAR ACTIVITIES

Workshop on Role of Calcium in Cell Signalling - 2020

Hands-on Training Workshop in Enzyme Kinetics - 2020

EXTRA-CURRICULAR ACTIVITIES

Athletics and fitness activities Badminton and other indoor/outdoor sports Drawing, painting and craft projects, board and strategic games

LANGUAGES

English (Proficient)

Kannada (Proficient)

Tamil (Good Working Knowledge)